Temporary Figure Gallery for CDA Paper Planning

Important. Every figure in this document uses *synthetic illustrative values* created only to preview layout, visual style, and narrative structure. These are not real experimental claims and should not be copied into the paper as results.

Goal

This temporary document previews what the empirical and diagnostic figure program could look like if the paper presents a broad domain-generalization story against common baselines such as ERM, SWAD, and DIWA while emphasizing the CDA-based method family.

Main-Paper Candidates

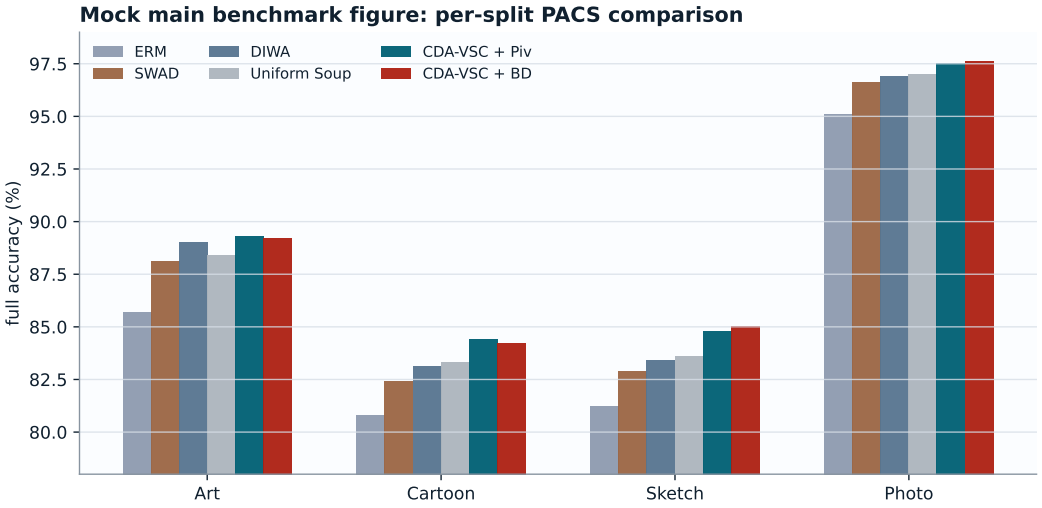

Figure 1: **Mock main benchmark comparison.** Grouped bars compare ERM, SWAD, DIWA, a uniform soup baseline, and the two CDA variants on each PACS target split. This is the most conventional "main result" figure and gives the reader an immediate sense of split-wise behavior and whether the proposed method is consistently competitive.

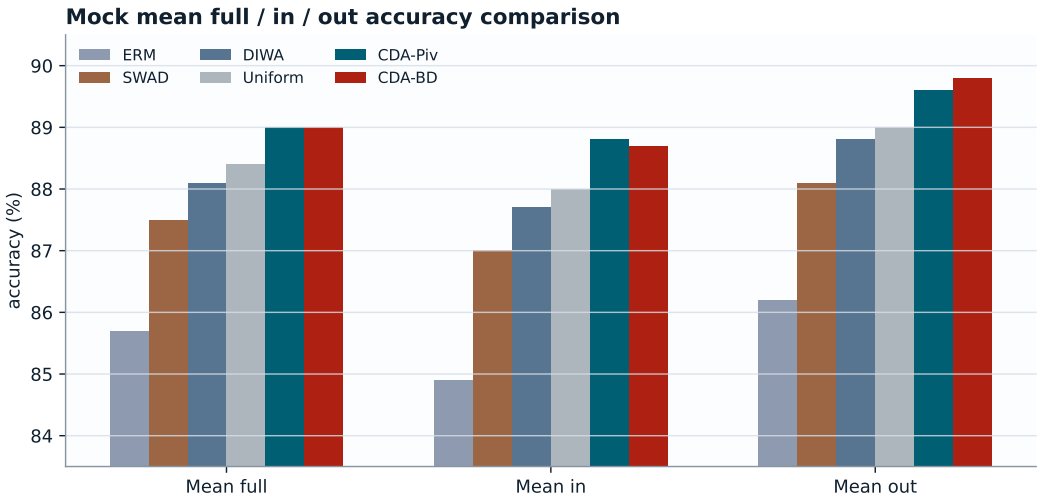

Figure 2: **Mock average full / in / out accuracy figure.** This figure would pair naturally with a table in the paper and makes it easy to show whether a method improves only pooled accuracy or also improves the out partition that is most relevant to the CDA narrative.

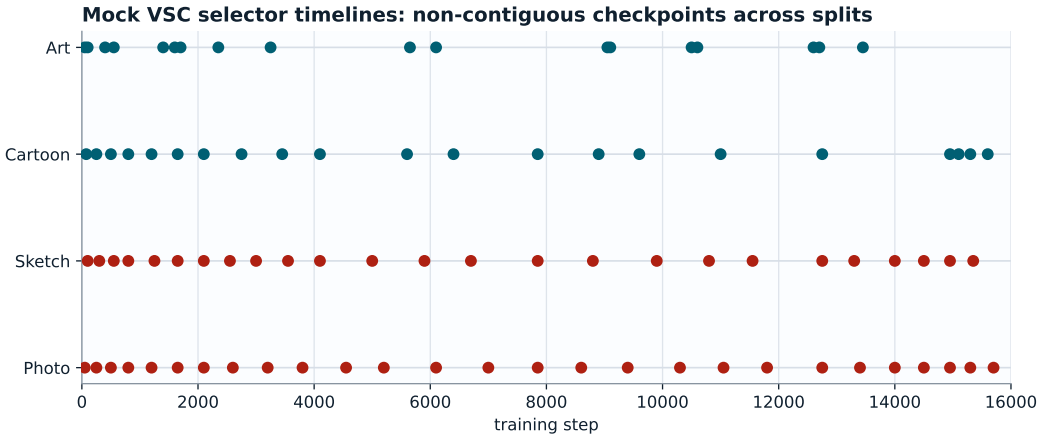

Figure 3: **Mock non-contiguous checkpoint timeline.** Each row shows the retained checkpoints on one target split. This figure is designed to support the claim that the selected family need not be a contiguous late-training window in order to be averageable and useful.

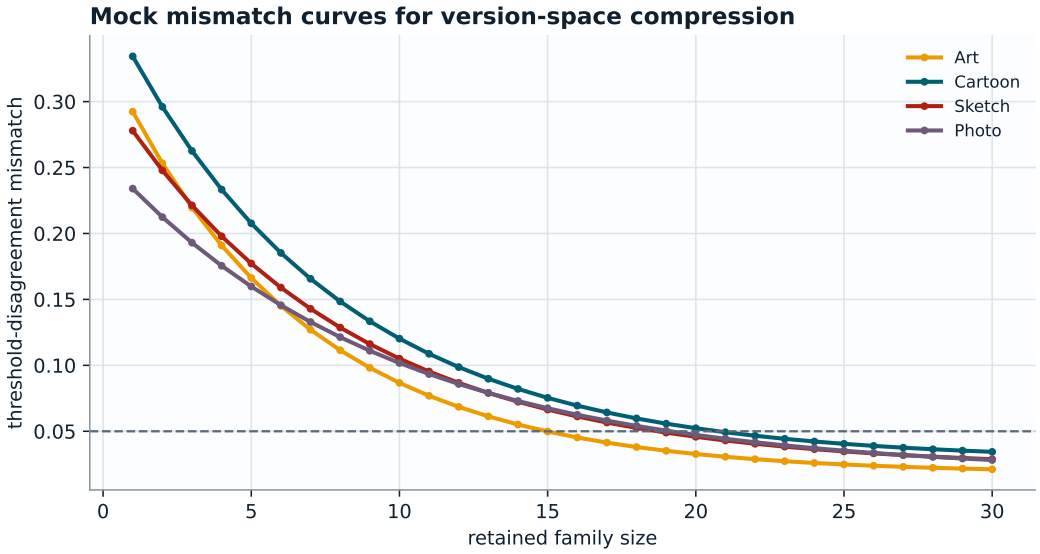

Figure 4: **Mock VSC mismatch curves.** The horizontal axis is retained family size and the vertical axis is disagreement mismatch to the full checkpoint bank. This is the cleanest selector-specific diagnostic because it shows whether added checkpoints are genuinely preserving the bank's source-support structure rather than merely increasing size.

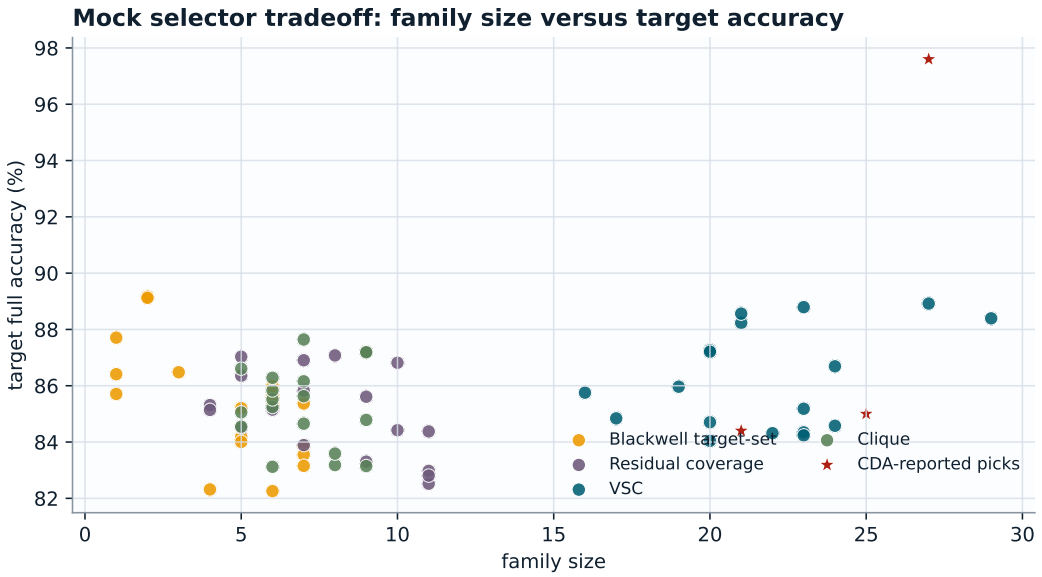

Figure 5: **Mock selector tradeoff plot.** Each point is a selector instance; the horizontal axis is family size and the vertical axis is target accuracy. This figure is useful for showing that over-compression to tiny families can kill the weighting story even when a selector looks strong on a single split.

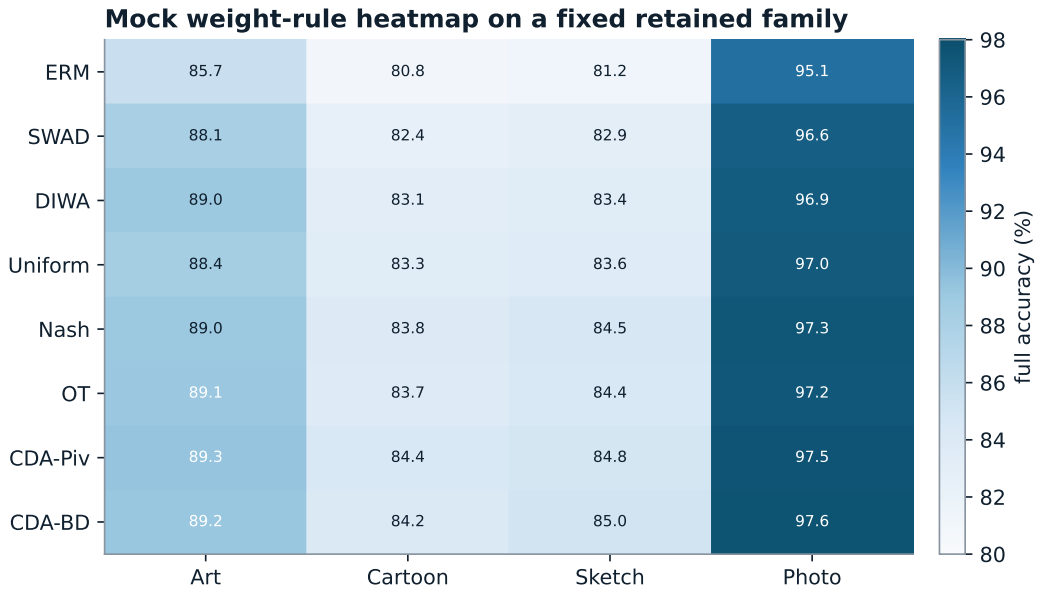

Figure 6: **Mock fixed-selector heatmap over weight rules.** The heatmap visualizes how different weighting rules behave once the selector is frozen. This figure helps distinguish whether the observed gains are uniquely tied to the proposed CDA-derived rules or whether several weighting strategies occupy the same performance band.

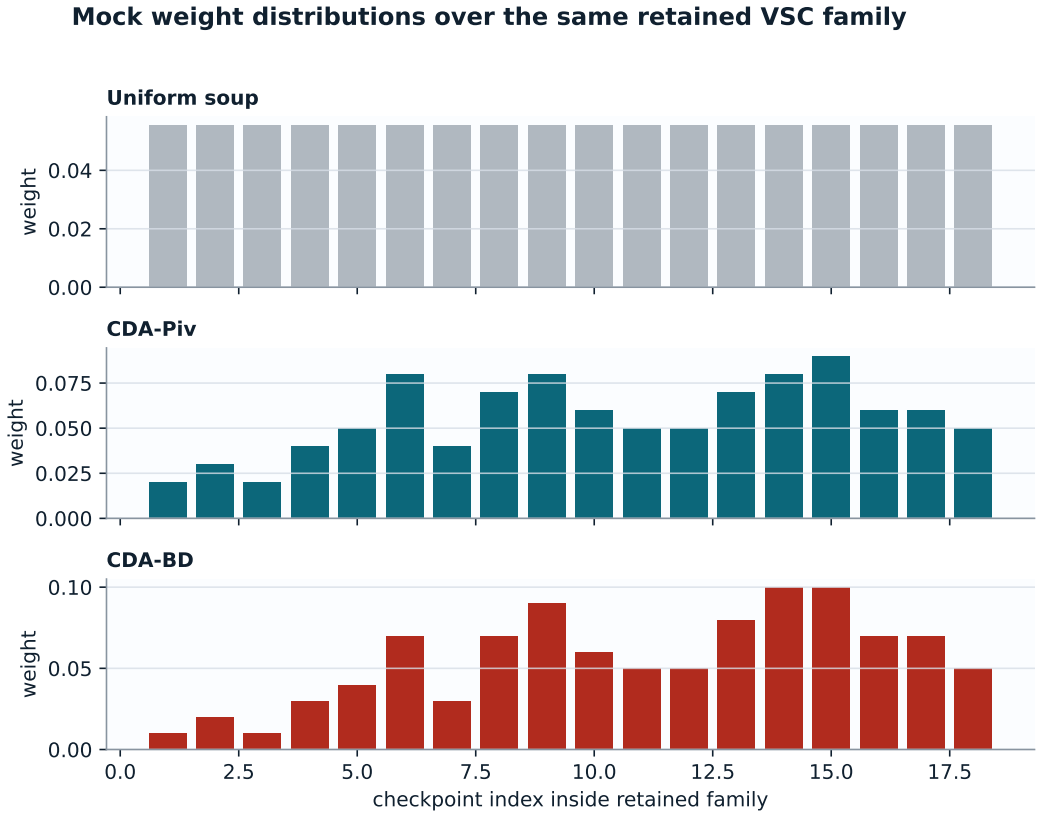

Figure 7: **Mock weight profiles on the same retained family.** The three panels compare uniform weighting, the pivotality-based rule, and the Blackwell-dual rule on the same checkpoint family. This is one of the most concrete ways to show that the proposed methods are not black boxes: they visibly reshape mass over the family rather than simply returning something uniform.

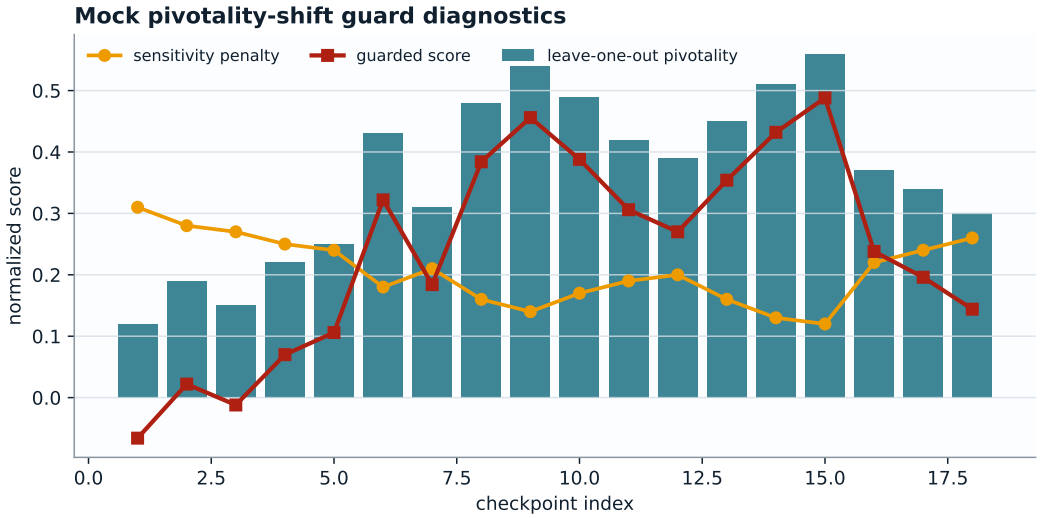

Figure 8: **Mock pivotality-shift guard diagnostics.** Bars show leave-one-out pivotality, the line shows the sensitivity penalty, and the final guarded score is overlaid. This figure would make the `pivotality_shift_guard` mechanism much easier for readers to understand than equations alone.

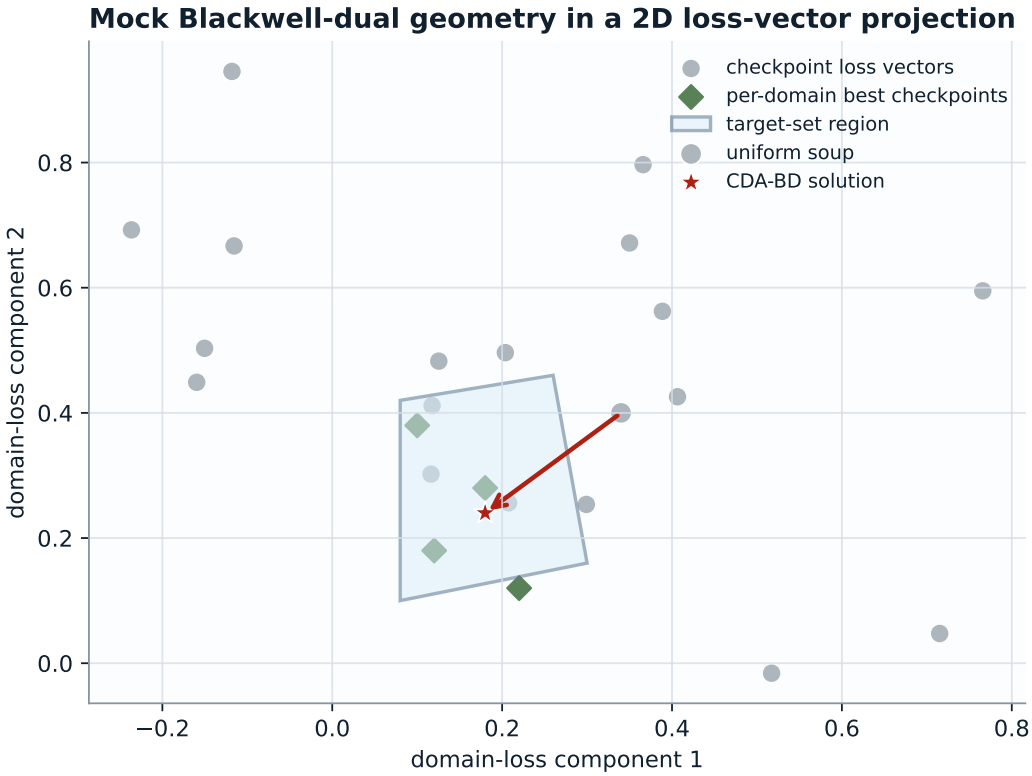

Figure 9: **Mock Blackwell-dual geometry.** Checkpoint domain-loss vectors are shown in a low-dimensional projection, together with a target-set region and the movement from a uniform soup toward the Blackwell-dual solution. This is the most intuitive empirical figure for the geometry of the dual weighting rule.

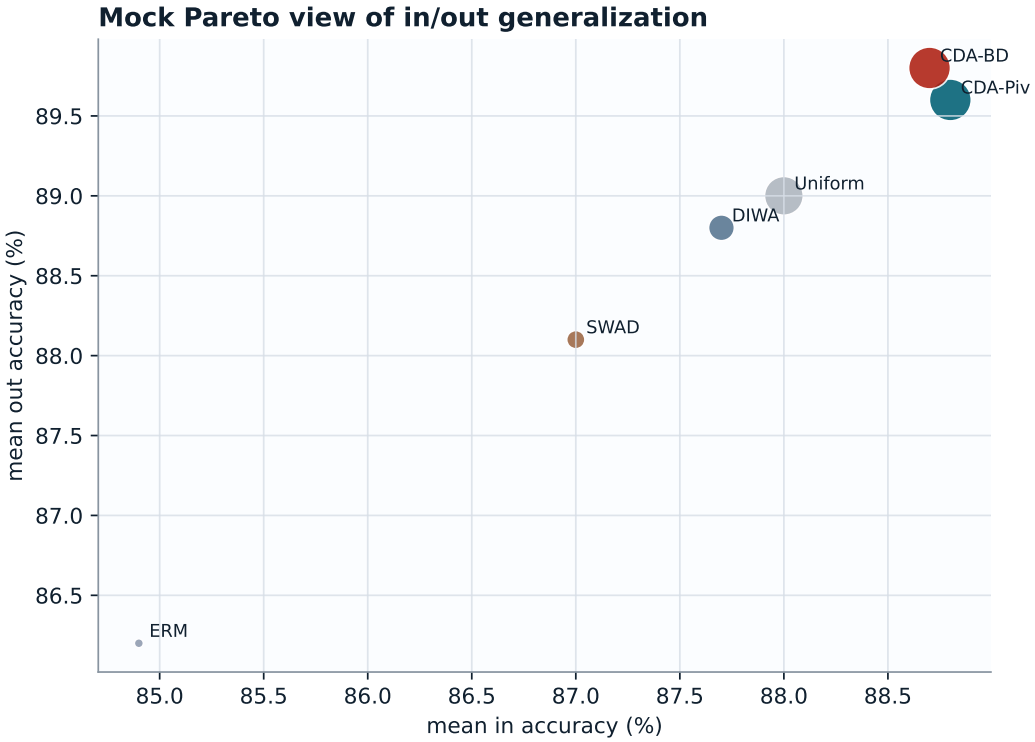

Figure 10: **Mock in-vs-out Pareto plot.** Each method is shown as a point in the in/out plane, with point size representing retained-family scale. This figure is useful when the paper wants to show that a method is not winning only by sacrificing one partition for the other.

Theory-to-Empirics Bridge Figures

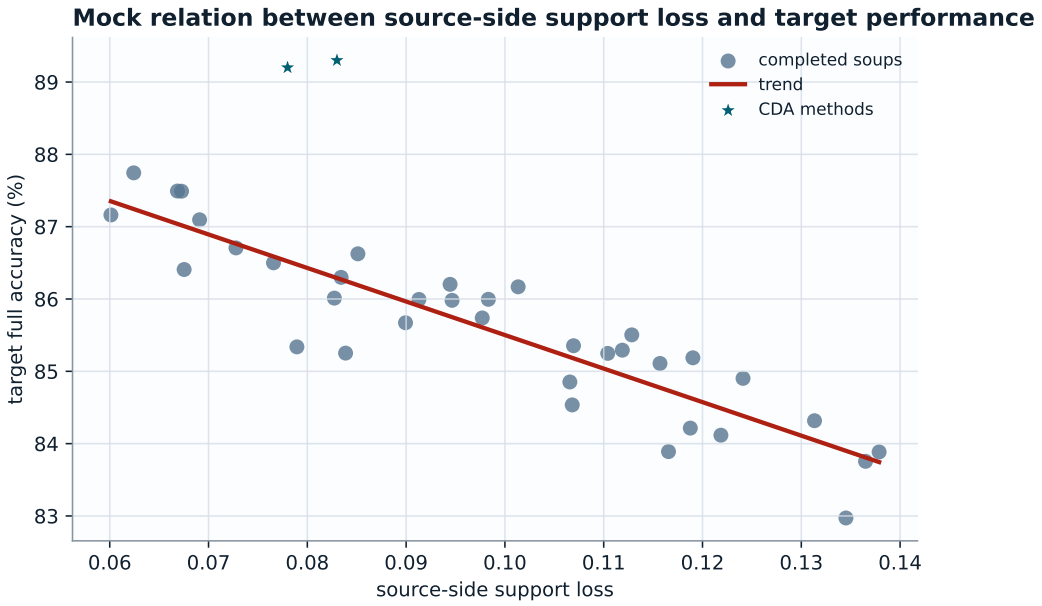

Figure 11: **Mock source-side proxy versus target accuracy.** This plot is designed to test whether a source-only diagnostic such as support loss or a CDA proxy actually predicts target behavior. If the paper wants its theory to do explanatory work, a figure of this kind is important.

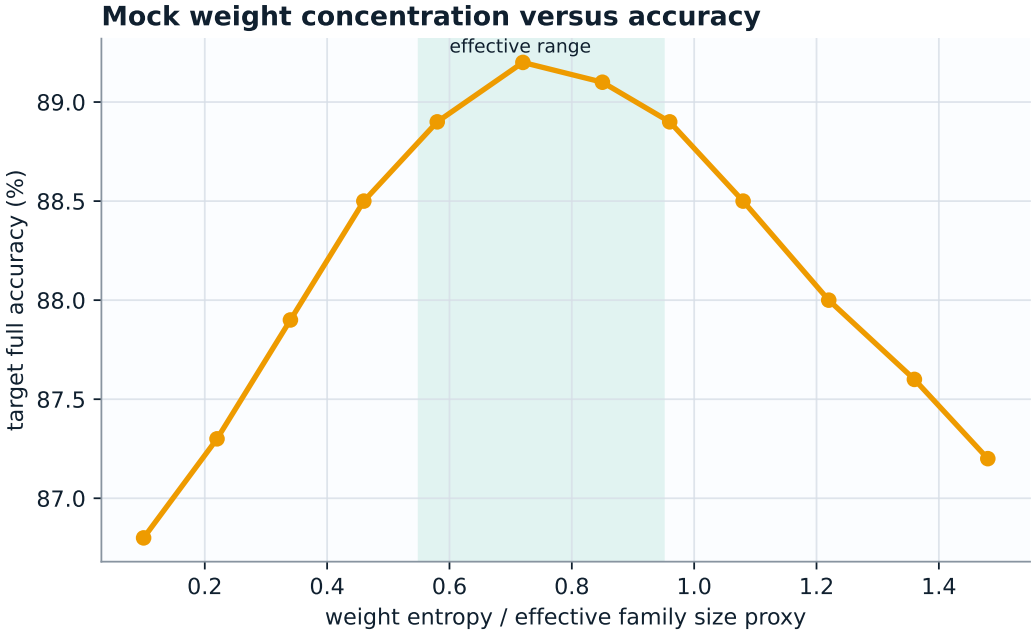

Figure 12: **Mock weight concentration versus accuracy.** The horizontal axis reflects how concentrated the final weights are. This helps show whether the method improves by learning a meaningful mixture or by collapsing onto a near-singleton solution.

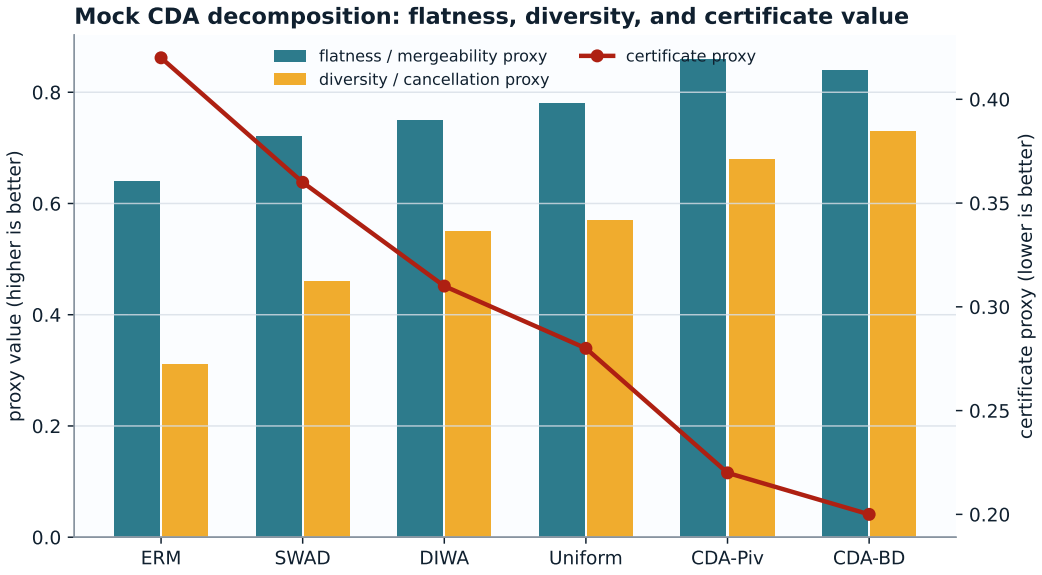

Figure 13: **Mock CDA decomposition plot.** The bars represent flatness / mergeability and diversity / cancellation proxies, while the line shows a certificate proxy. This figure would visually support the unifying claim that apparently different successful heuristics are reducing different parts of the same source-side criterion.

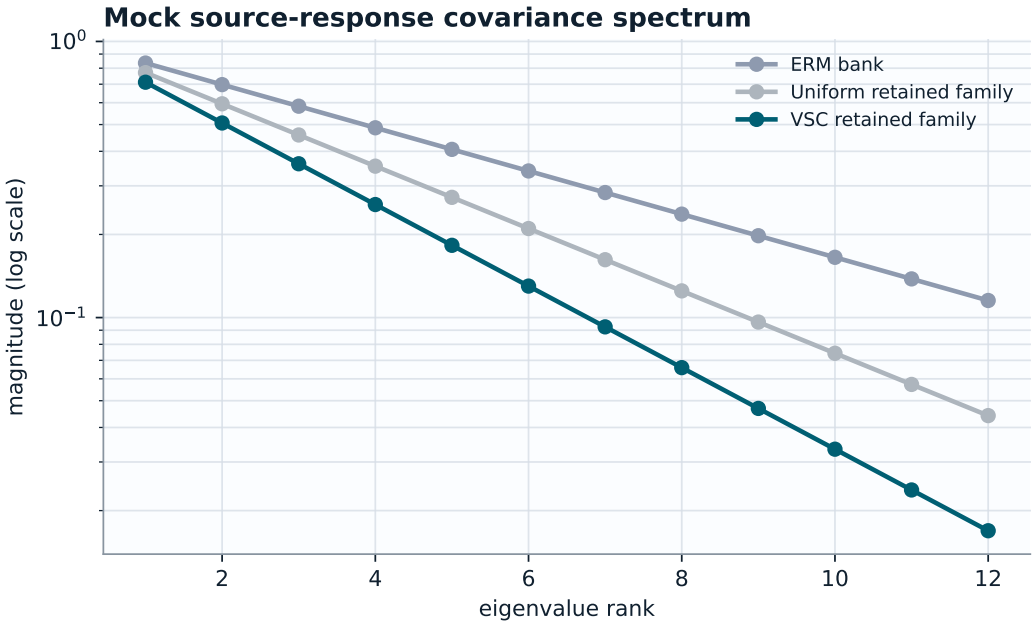

Figure 14: **Mock covariance spectrum diagnostic.** This scree-style figure shows how much domain-response variation remains after family compression. It is a natural visual for the claim that version-space compression preserves the important low-dimensional disagreement geometry of the full bank.

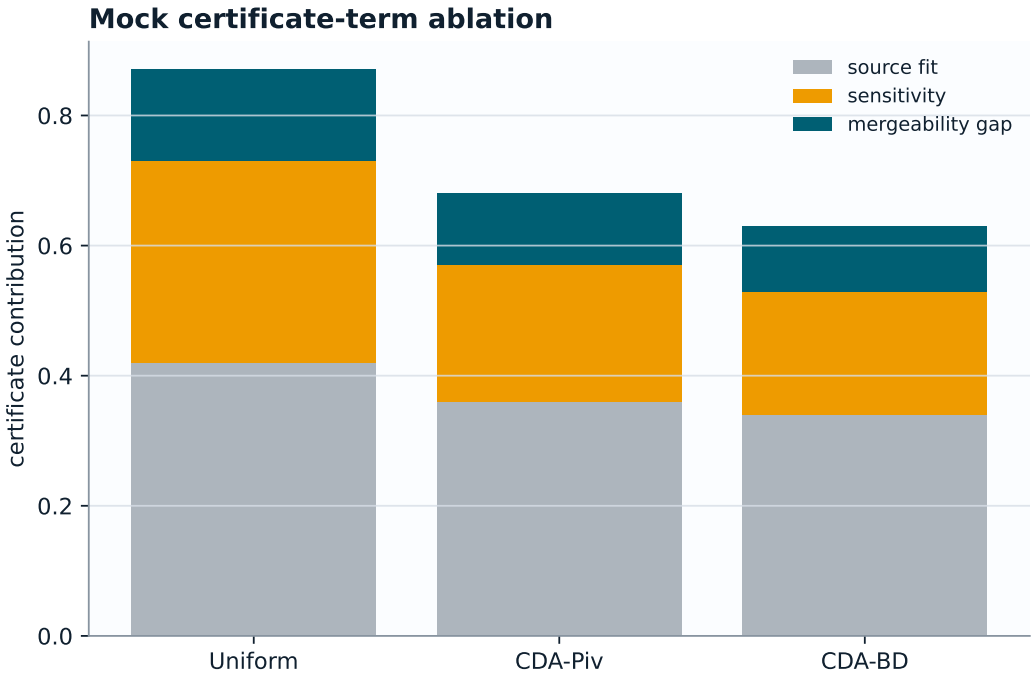

Figure 15: **Mock certificate-term ablation.** Each stacked bar decomposes a source-supported certificate into source fit, mixture sensitivity, and mergeability gap. This is a strong figure if the paper wants to demonstrate that the proposed methods help for the reason the theory predicts, not just because they score well.

Appendix-Style Diagnostics

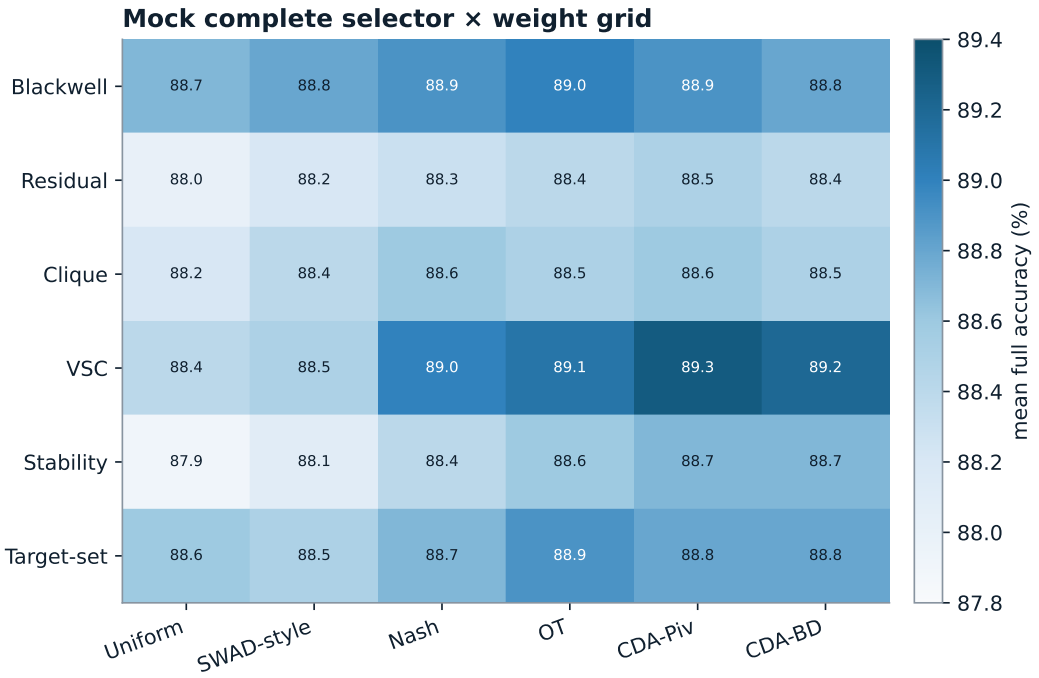

Figure 16: **Mock full selector-by-weight heatmap.** This appendix-style figure visualizes the broader search space after the main method is chosen. It is useful for showing that the final method was selected from a landscape with meaningful structure rather than from a flat sea of near-identical results.

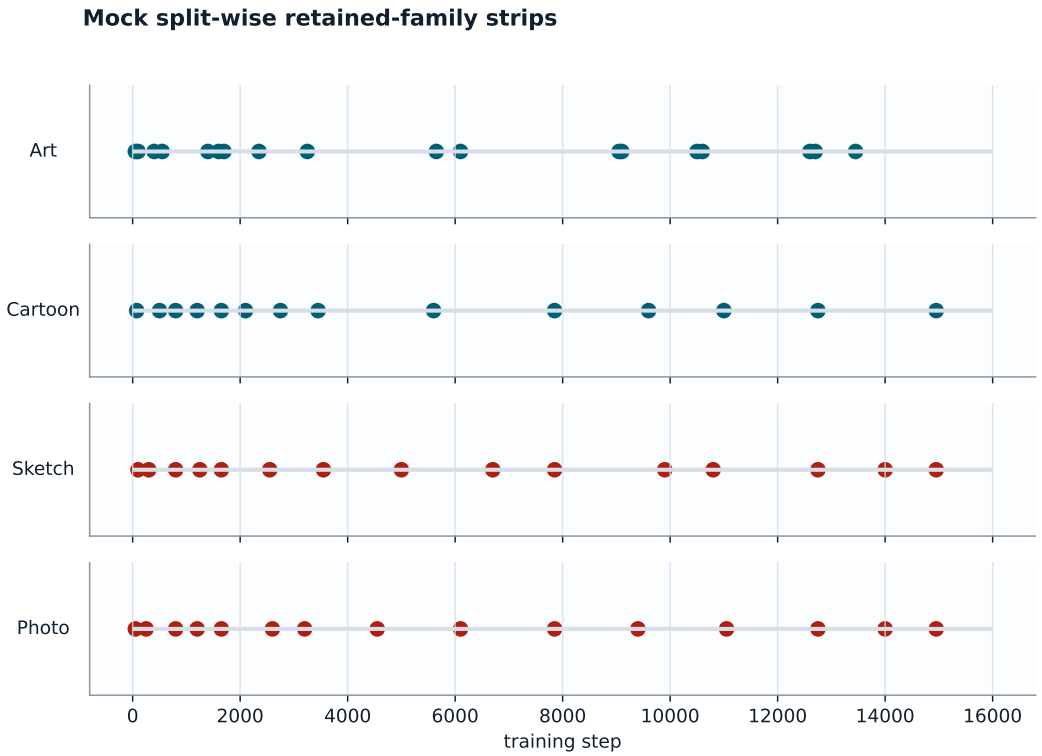

Figure 17: **Mock split-wise retained-family strips.** This is a compact appendix-friendly way to show exactly which checkpoints were retained on each split without forcing the reader to parse a long list of step indices in the main text.

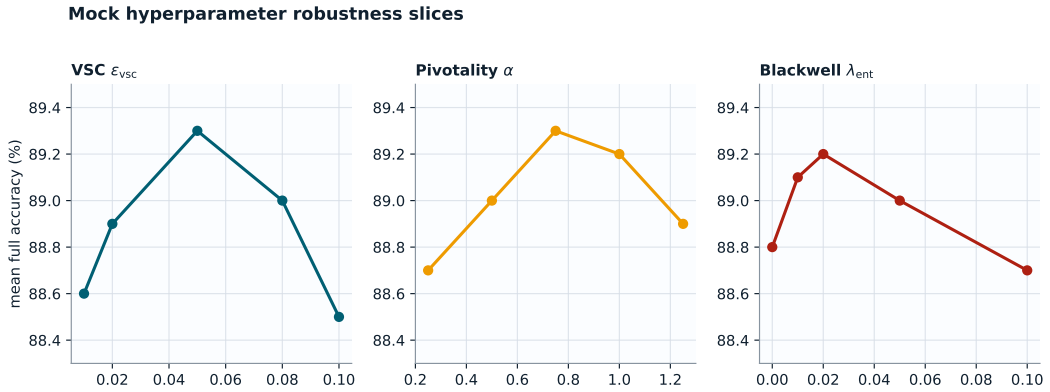

Figure 18: **Mock hyperparameter robustness slices.** Each panel shows a one-dimensional slice through a selector or weighting-rule hyperparameter. This kind of figure helps distinguish a method that is scientifically stable from one that only works at a single lucky setting.

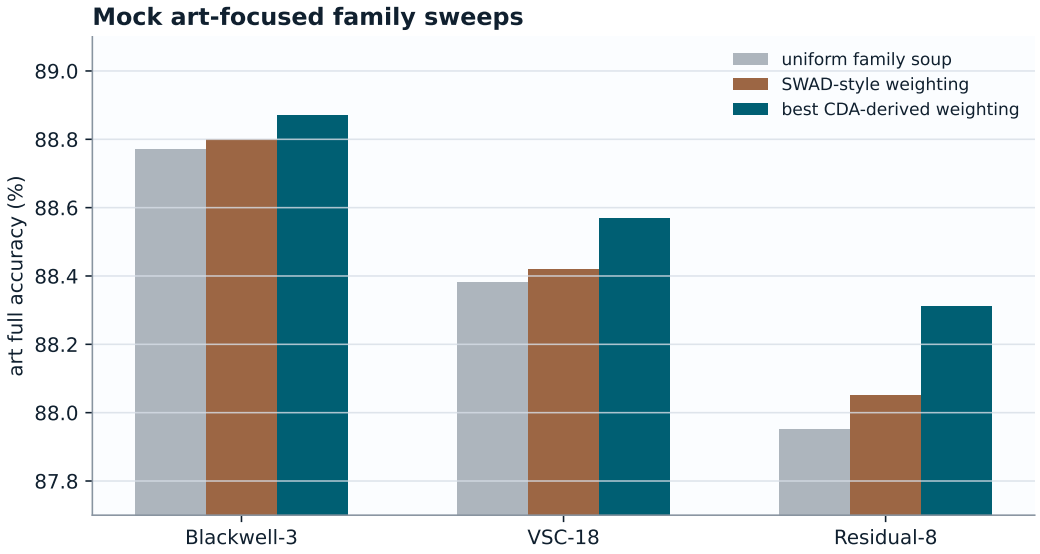

Figure 19: **Mock art-focused family sweeps.** This figure compares uniform family soups, SWAD-style weighting, and the best CDA-derived weighting across several art-focused families. It is a clean way to show that the benefit of nonuniform weighting depends on the geometry and size of the retained family.

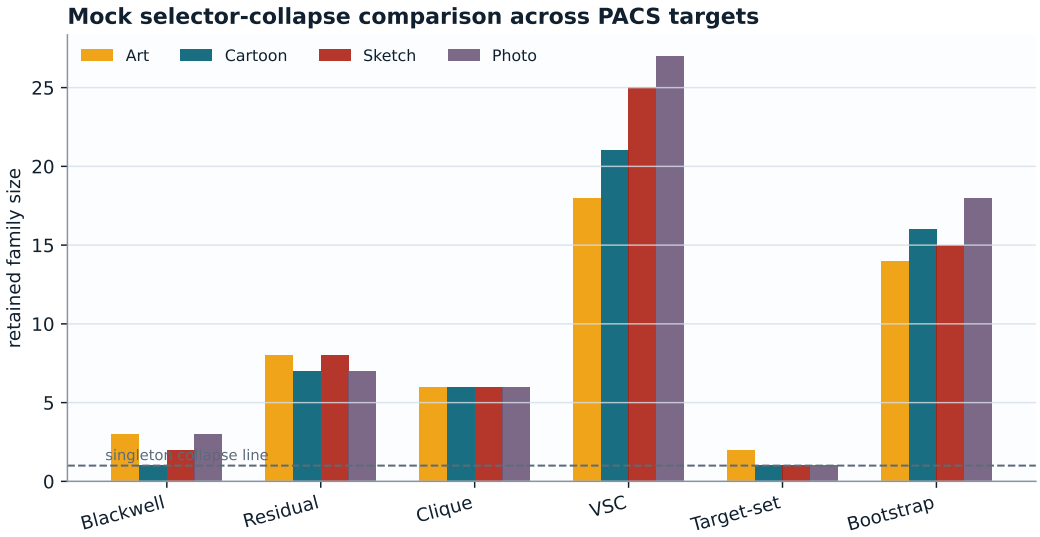

Figure 20: **Mock selector-collapse comparison.** Bars show retained family size across PACS target splits for several selector families. This figure directly supports the paper's argument that a selector can fail by collapsing too aggressively, even when it appears strong on one split.

How To Use This Preview

- Promote the strongest 6–8 figures into the main paper.
- Move the broader search-space plots into the appendix.
- Replace the synthetic numbers with real exports from the results ledger and replay artifacts once the final figure list is chosen.